MEHDI **EIDI**

M.Sc. Student & Researcher in Software Engineering
Department of Computer Engineering, Sharif University of Technology, Tehran, Iran
eidim@acm.org | Homepage | Google Scholar | GitHub | ORCID | LinkedIn

Research Interests

Model-Driven Engineering • AI for Software Engineering • Empirical Software Engineering • Automated Software Engineering • Software Architecture.

Education

M.Sc. in Software Engineering, Sharif University of Technology — Sep 2024 - Feb 2027 Exp. — Tehran, Iran

- **GPA:** 19.5/20 | **Ranked 1st by GPA**
- **Thesis:** AI-Assisted Model-Driven Methodology for Serverless Software Development | Website
- **Supervisor:** Dr. Raman Ramsin
- **Selected coursework:** Software Architecture (20/20), Patterns in Software Engineering (20/20), Software Development Methodologies (20/20), Requirements Engineering (19.2/20), Software Testing (19/20).

B.Sc. in Computer Engineering, Urmia University — Sep 2019 - Aug 2023 — Urmia, Iran

- **GPA:** 18.73/20 | **Ranked 3rd by GPA**
- **Industry-collaborative capstone project:** Designed, implemented, and deployed a microservices-based call-center management system that remains in active production use.
- **Selected coursework:** Machine Learning (20/20), Software Engineering (20/20), Systems Analysis & Design (20/20), Data Mining (20/20), Cloud Computing (20/20).

Research Experience

M.Sc. Researcher, Methodology Engineering Lab — Dec 2024 - Present

- **Project:** AI-Assisted Model-Driven Methodology for Serverless Software Development Website | GitHub
- **Research Focus:** Investigating how model-driven engineering and LLMs can support the systematic development of serverless software from requirements to implementation artifacts.
- **Methodology Contribution:** Developed the MODRISS methodology that integrates a development process, a modeling framework, and an LLM-based modeling assistant.
- **Language Engineering:** Designed three DSMLs for modeling the business domain, generic serverless architectures, and the AWS serverless ecosystem.
- **Model-Driven Engineering:** Implemented automated CIM→PIM→PSM model transformations and PSM→project code generation, enabling progressive refinement from domain models to implementation artifacts.
- **Method Engineering:** Engineered a serverless development process spanning development, delivery, operations, and maintenance.
- **LLM4MDE:** Developed an LLM-based modeling assistant that creates and evolves structured models from natural-language requirements.
- **Evaluation:** Evaluated MODRISS through industrial case studies and a criteria-based assessment.

Publications

[1] **Eidi, M.**, & Ramsin, R. (2026). Model-Driven Approaches for Serverless Software Development: Evaluation and Future Directions. *Proceedings of the 14th International Conference on Model-Based Software and Systems Engineering (MODELSWARD'26)*. doi:10.5220/0014634200004058

[2] **Eidi, M.**, & Ramsin, R. MODRISS: AI-Assisted Model-Driven Methodology for Serverless Software Development. *(Manuscript in preparation)*

Conference Presentations

Presented paper **[1]** at the *14th International Conference on Model-Based Software and Systems Engineering (MODELSWARD'26)*, Marbella, Spain, March 9, 2026.

Academic Distinctions

- **Ranked 63rd of ~20,000 candidates** in the National M.Sc. Entrance Examination (**top 0.3%**).
- **Ranked 1st by GPA**, M.Sc. Software Engineering, Sharif University of Technology.
- **Ranked 3rd by GPA**, B.Sc. Computer Engineering, Urmia University.

Teaching Experience

Teaching Assistant, Sharif University of Technology

- Agile Software Development (2 semesters) — Fall 2024, Fall 2025
- Object-Oriented Systems Design — Spring 2025

Teaching Assistant, Urmia University

- Data Structures & Algorithms — Spring 2022
- Advanced Programming — Fall 2022
- Fundamentals of Programming — Fall 2021

Go Workshop Instructor, Urmia University — Spring 2021

- Taught Go programming to university students through five hands-on workshop sessions.

Industry Experience

Backend Software Engineer, Imenaria — Jun 2022 - Sep 2024

- Designed and implemented RESTful and gRPC services for a video-on-demand platform and engineered a scalable video-transcoding subsystem for high-volume media processing.
- Developed typo-tolerant search and executed a large data migration from MongoDB to Postgres.
- Built an automated NGINX configuration generator and a real-time chat subsystem.

Software Architect & Backend Engineer, HUB — Feb 2023 - Jul 2023

B.Sc. Capstone Project in Collaboration with Industry

- Designed, implemented, and deployed a microservices-based call-center management system.
- Performed architectural and detailed design using UML.
- Optimized a critical API through batch processing, reducing latency from ~10 seconds to 200 ms.
- Established a CI/CD pipeline and deployment infrastructure and containerized services.

Open-Source Projects

Carify: Machine learning project predicting used-car prices based on vehicle condition. GitHub
Offliner: Efficient concurrent crawler, mirroring websites offline. GitHub
Tunelo: Proxy, tunneling UDP/WireGuard over WebSocket or TCP. GitHub
GoShell: Unix-like shell with pipelines, I/O redirection, and background jobs. GitHub

Technical Skills

Model-Driven Engineering: EMF, Eclipse Epsilon, UML, OCL, BPMN, SPEM
Programming: Java, Go, Python
DevOps & Cloud: Docker, Git, Linux, AWS
Data: PostgreSQL, Redis, SQL, MongoDB, RabbitMQ
Design: Design & Architectural Patterns, Microservices, Event-Driven Architecture, Refactoring

Languages

English: C1 (Full Professional Proficiency) **Turkish:** Native **Azeri:** Native **Persian:** Native

References

Raman Ramsin, Ph.D. Email | Homepage
Associate Professor, Department of Computer Engineering, Sharif University of Technology

Jafar Habibi, Ph.D. Email | Homepage
Full Professor, Department of Computer Engineering, Sharif University of Technology

Seyed-Hassan Mirian-Hosseinabadi, Ph.D. Email | Homepage
Associate Professor, Department of Computer Engineering, Sharif University of Technology